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Volume 2 - Applications Guide: Appendix B - Programme Analysis Case Studies

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Appendix B1 - Programme Analysis

Case Study 1

1 Western Province rehabilitation programme – Life Cycle Analysis Method

1.1 Introduction

This Case Study demonstrates the application of Programme Analysis to the production of a rehabilitation programme for a road network in the Western Province of a country. The study demonstrates the Life Cycle Analysis method described in Chapter B2.

The objective here is to prepare a prioritised list road projects from a selected group of road sections that have been identified to be in poor condition. In practice, this would follow some condition survey of the road network and the application of pavement condition trigger levels or thresholds to identify those sections in need of some form of rehabilitation.

The Life Cycle analysis method used for this Case Study is the same as that used in Project Analysis and described in Chapter B1.

To demonstrate this Programme analysis method, the Case Study is presented in the following steps:

- Step 1 : Overview of the case study.
- Step 2 : Review of the case study input data.
- Step 3 : Run programme analysis with the life-cycle option
- Step 4: Examine the results of the work programme

1.2 Overview of the case study data

The input data for this Case Study is labelled as:

- Western Province Rehabilitation (Life cycle analysis).

When a Programme Analysis study is created, the following data sets must be defined:

- Road network containing the candidate road sections for periodic maintenance and/or improvement.
- Vehicle fleet defining the characteristics of typical vehicles commonly found on the road network.
- The average annual daily traffic (AADT) on each road section.
- Maintenance and/or improvement standards to be assigned to each candidate road section.

The above data will usually be pre-defined in HDM-4 before a Programme Analysis study is created. The following data managers are used for this:

- **Road Network manager** - to create and edit the road section data

- **Vehicle Fleet manager** - to create and edit vehicle characteristics
- **Maintenance and Improvement Standards manager**

To review the case study data, open the Programmes folder and select the **Western Province Rehabilitation (Life Cycle Analysis)** study as illustrated in Figure B1.1.

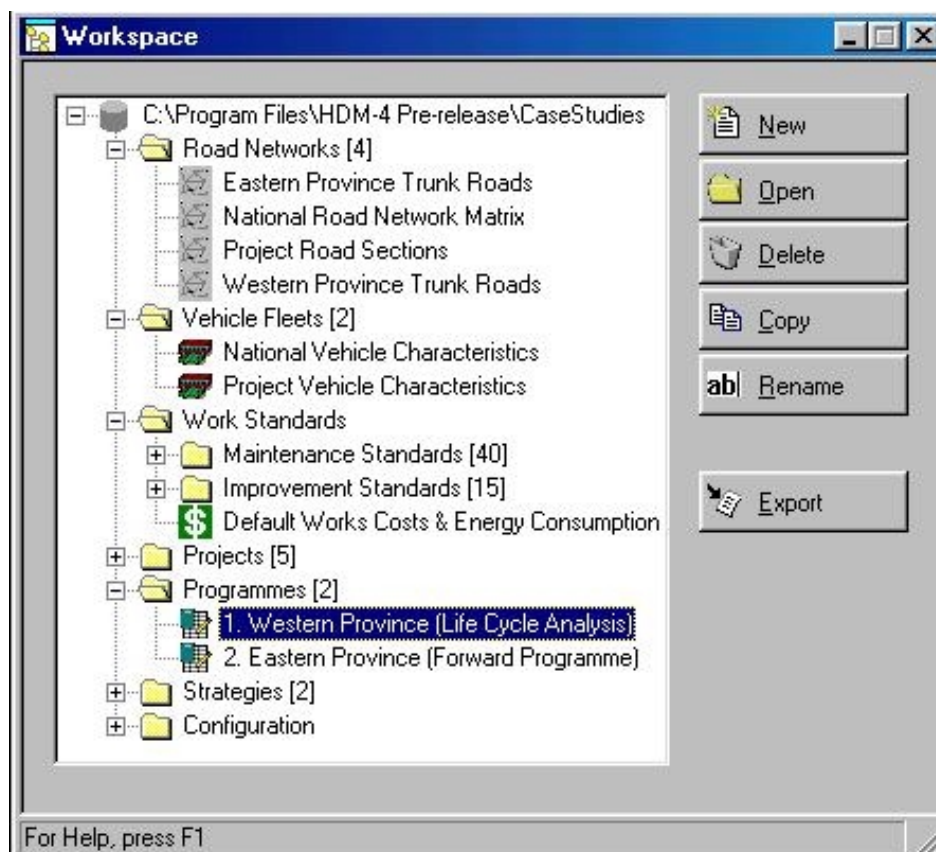


Figure B1.1 Workspace screen

1.3 Input data

The data used for this Case Study has already been created and therefore the procedure described below does not include the creation of a new study.

The procedure involves defining details of the programme analysis to be carried out, assignment of work standards, generation of the work programme, optimisation under budget constraints, and finally generation of reports. This procedure should normally be followed sequentially, although iteration between the above tasks can be done in order to review previously specified data.

The data is arranged in groups (corresponding to the tasks) and, within these, the data are presented in different **Tabs** as summarised below:

Define Programme Details

- **General Tab**
- **Select Sections Tab**

- **Select Vehicles** Tab
- **Define Traffic** Tab

Assign Standards

- **Specify Alternatives** Tab

Generate Programme

- **Perform Run** Tab
- **Unconstrained Programme** Tab

Perform Budget Optimisation

- **Define Budget** Tab
- **Optimised Programme** Tab

Generate Reports

- **Reports Tree**

1.4 Define Programme Details

Programme: 1. Western Province (Life Cycle Analysis)

Define Programme Details | Specify Standards Assignments | Generate Programme | Perform Budget Optimisation | Generate Reports | Save | Close

General | Select Sections | Select Vehicles | Define Normal Traffic

Study Description:

Type of analysis: ☒ Life-cycle ☐ Multi-Year Forward Programme

Start year: Duration: years

Road Network:

Vehicle Fleet:

Currencies

Fleet: US Dollar × = output currency

Works: US Dollar × = output currency

Output: Discount rate:

For Help, press F1

1.4.1 General

The **General** Tab can be used to define the study description, analysis type, analysis period (start year and duration), the currency conversion rates and the discount rate to be used in the economic analysis. This Tab also shows the selected **Road Network** and **Vehicle Fleet**.

This Case Study involves Life Cycle analysis carried out over a fifteen year analysis period starting from 2000 to 2014.

1.4.2 Select Sections

It is assumed that the long list of candidate road sections from Western Province was drawn up following some pavement condition survey. The list would normally contain only those road sections deemed to require some form of periodic maintenance or rehabilitation during the next budget period. Consequently, all road sections have been selected for the analysis.

However, if the road network had contained details of all road sections in Western Province, it is possible to select only those that should be included in the work programme for the next budget period.

Alternatively, all road sections in the database could be analysed, particularly if this involves a long budget period.

It is recommended that work programmes should be prepared for short term budget periods, for example five years or less.

Details of each road section can be edited from within this Tab (press the **View/Edit Network** button).

1.4.3 Select Vehicles

A typical analysis using HDM-4 will involve the calculation of road user costs, primarily in the form of vehicle operating costs, and travel time costs.

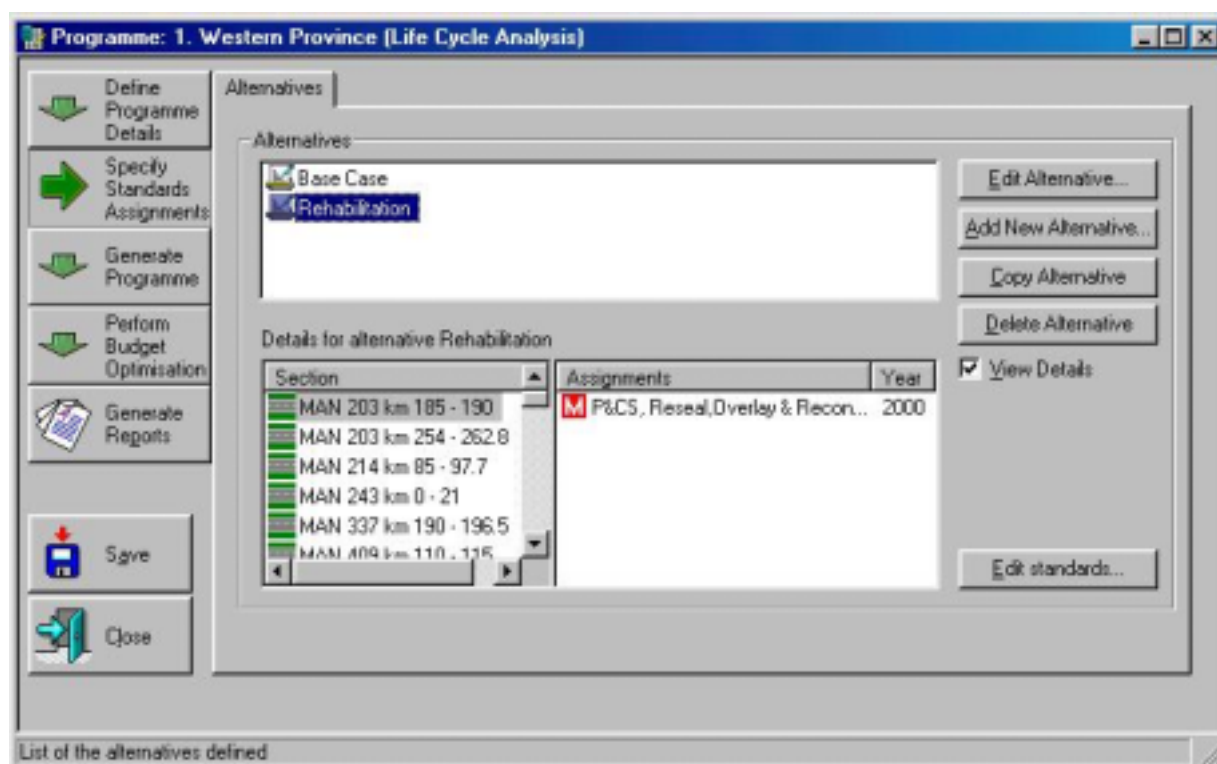
This Tab can be used to select which vehicle types will be included in the analyses, from amongst those specified in the **Vehicle Fleet** database.

It is also possible to review or amend details of the vehicle types from within this Tab.

1.4.4 Define Normal Traffic

The AADT observed on each of the road sections will have been specified together with details of the road sections. This Tab specifies the traffic composition and annual growth rates that are applicable on each road section for each vehicle type in this analysis. Note that several traffic growth periods may be specified, for example, 5% annual growth between 2000 – 2004, followed by 3.5% annual growth from 2005 onwards. In addition, the traffic details can be copied from one road section to others, thereby permitting each road section to have a different road set of traffic growth characteristics, if required.

1.5 Specify Standards Assignments



1.5.1 Alternatives

The Life Cycle analysis method requires a comparison to be made between a base case (**do minimum**) alternative against a project case (**do something**) alternative. Details of at least two alternatives can be specified in this Tab.

For this Case Study, two alternatives have been defined for each road section; a **base case** alternative and a **rehabilitation** alternative. The base case comprises routine pavement maintenance only (crack sealing and patching potholes only). The rehabilitation alternative includes routine maintenance, resealing, overlays and reconstruction for bituminous pavements. Details of the maintenance standards are summarised in Table B1.1.

1.5.2 Maintenance Standards

The **P&CS, resealing, overlay and reconstruction** standard, adopted for bituminous sections, includes the five maintenance works items described in Table B1.1. The standard has been assigned the short code: Brehab for “bituminous rehabilitation”.

Table B1.1
Details of Maintenance Standard (Brehab) assigned to bituminous road sections

Works Type	Works Activity	Intervention Criteria	Ranking *
Routine Maintenance	Patching	Potholes ≥ 10 per km	22
	Crack sealing	Transverse cracks ≥ 15 per km	22
Resealing	Single Surface Dressing	Total damaged area $\geq 30\%$ of pavement surface area	16
Overlays	Overlay dense graded asphalt	Roughness ≥ 5 IRI AND Cracking $\geq 5\%$	9
Reconstruction	Pavement reconstruction	Roughness ≥ 12 IRI	6

Note: * Ranking of road works from Chapter D2 of [Analytical Framework and Model Descriptions](#)

If more than one of these activities are triggered in a given year, the works item that is highest in the hierarchy of road works (that is, with lowest ranking value) will be applied.

Note that routine pavement works (for example, patching and crack sealing) have the same ranking, and therefore both can be applied in the same year. (The hierarchy is defined by the ranking of road works described in Chapter D2 of [Analytical Framework and Model Descriptions](#)).

Table B1.2
Details of alternatives for life-cycle analysis

Alternative	Surface Class	Maintenance Standard Description	Code	Effective from year	Maintenance Works	Code
Base Case	Bituminous	Patching and Crack Sealing	Broutine	2000	Pothole Patching Crack Sealing	PATPOT CRKSL
Rehabilitation	Bituminous	P&CS, Reseal, Overlay & Reconstruct	Brehab	2000	Pothole Patching * Crack Sealing * Reseal Overlay Reconstruct	PATPOT CRKSL RSL30 OV5IRI REC12

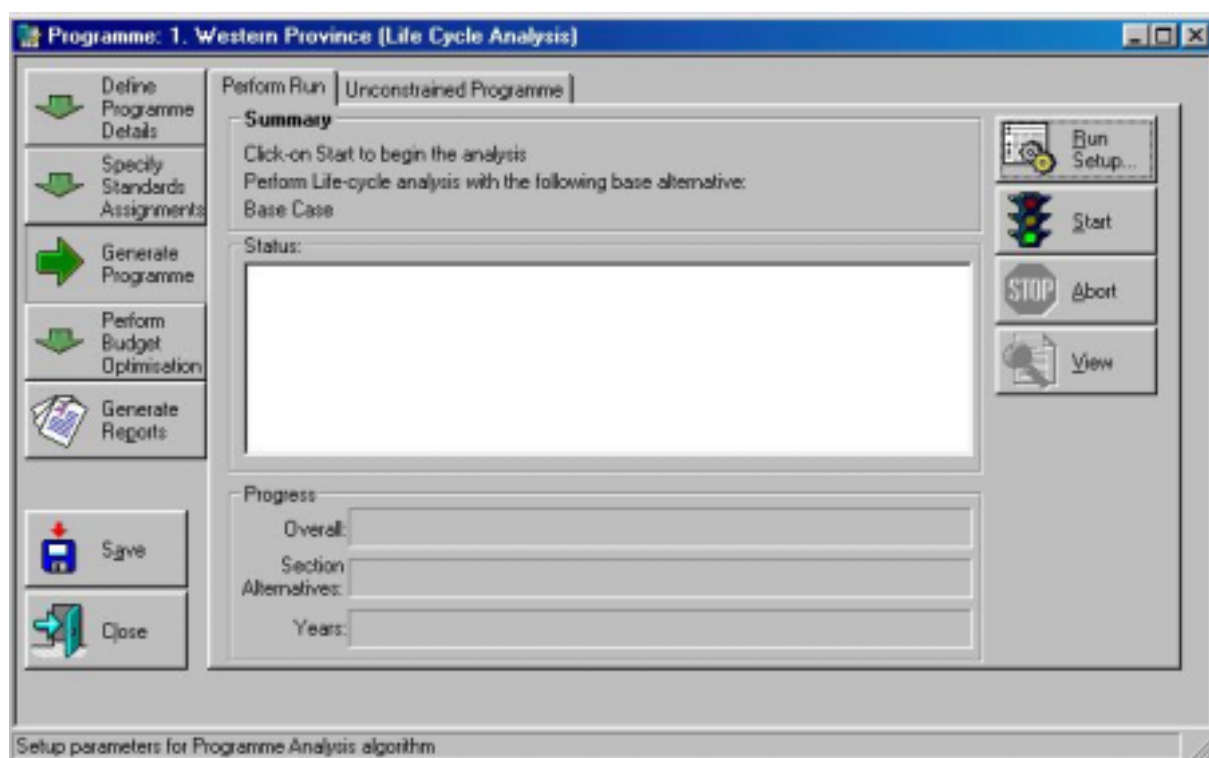
Notes:

- 1 Details of the intervention criteria for road work activities are given in Table B1.1.
- 2 The routine pavement works in the Rehabilitation alternative (identified by *) are identical to those in the Base Case.

1.6 Generate Programme

1.6.1 Perform Run

For life-cycle analysis, the **Base** alternative for economic analysis is specified in the **Run Setup** screen. This can be displayed by selecting the **Run Setup** button within the **Perform Run** Tab.



1.6.2 Work Programme

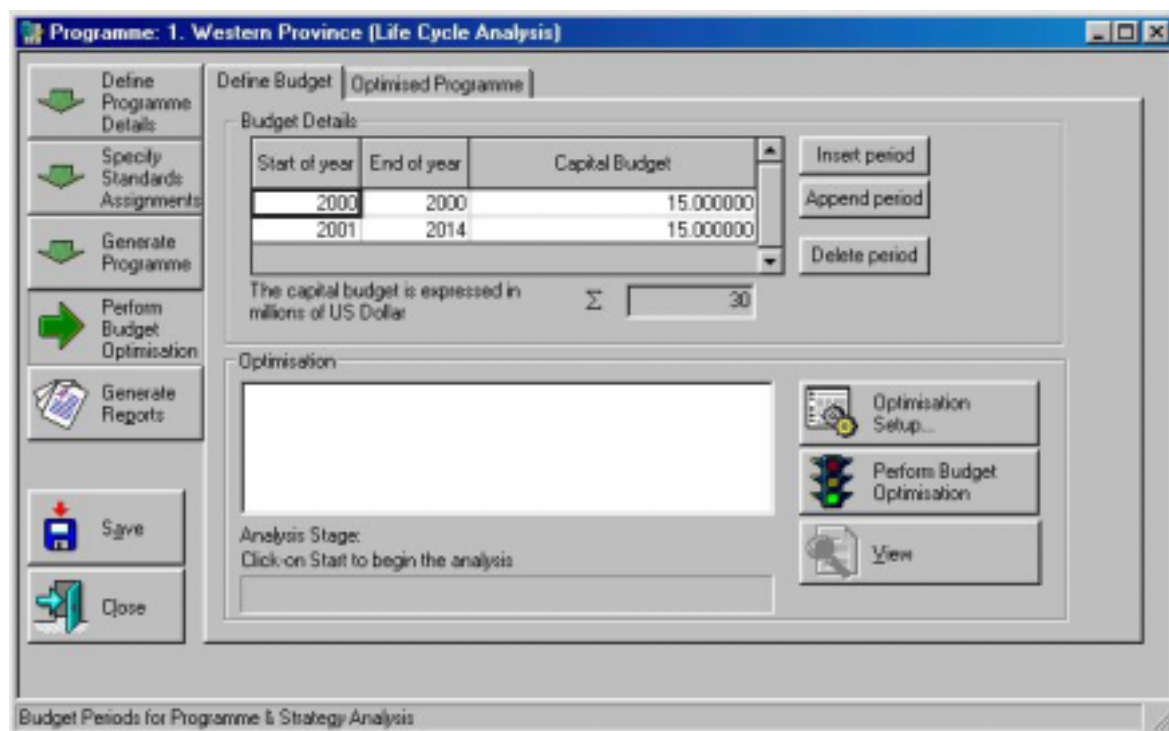
On completing **Run Setup**, press the **Start** button to commence the economic analysis, which produces an unconstrained work programme. This contains, for each road section, the alternative with the highest NPV (refer to Chapter B2). This represents an unconstrained programme, with the total financial cost given in the **cumulative cost** column.

The ideal budget required for periodic maintenance in year 2000 is approximately US\$ 22 million.

1.7 Perform Budget Optimisation

1.7.1 Define Budget

The budget optimisation provides a facility to select road sections that can be included within a specified budget in order to maximise the economic benefits. Given that the selected road sections from Western Province would require US\$ 22 million in year 2000 for periodic maintenance, it is now possible to select the optimal set of road sections if less than 100% of the required budget is available. Assume that the amount of money available for year 2000 (that is, the next budget period) has been set at US\$ 15 million.



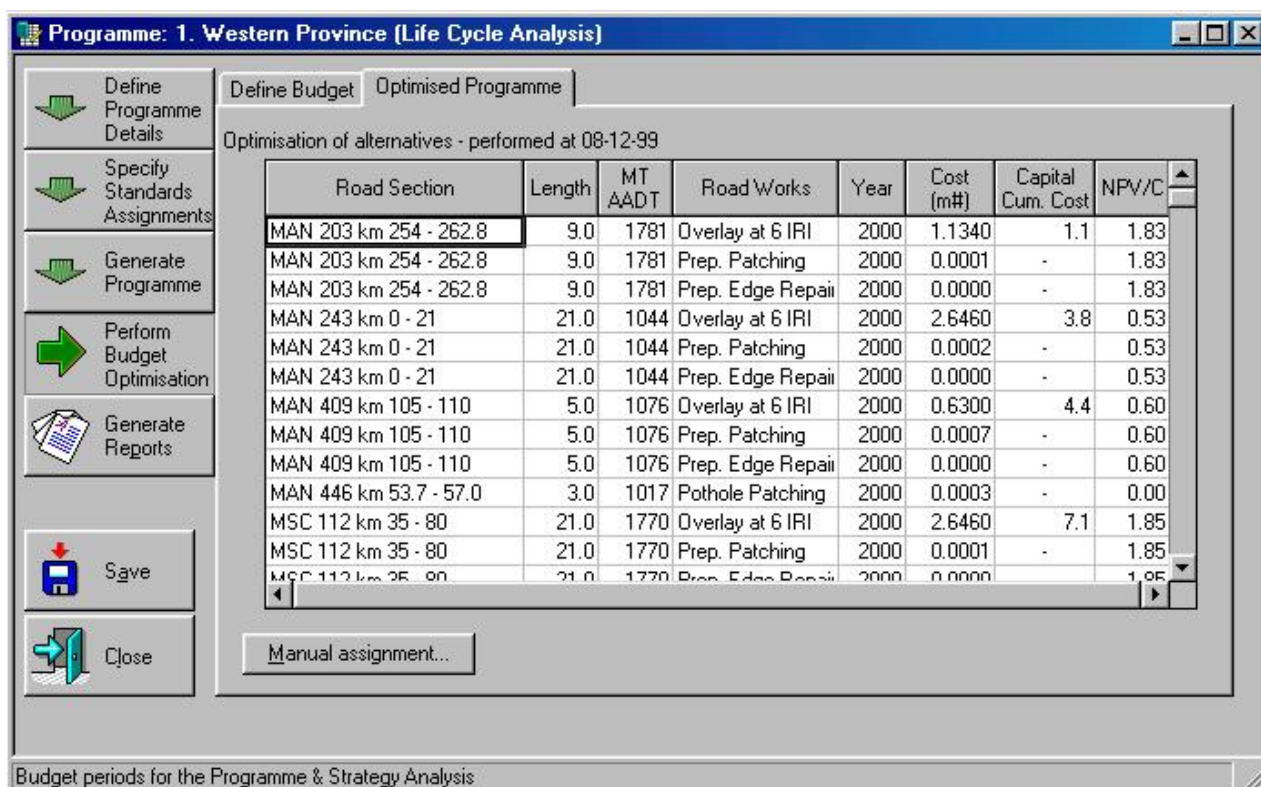
Details of the budget optimisation options are specified by pressing the **Optimisation Setup** button. For this case study, optimisation is by incremental benefit/cost ranking (see Chapter B2). The parameters to be specified are given below:

Minimum incremental value = 0.1

Efficiency frontier Zone = 75%

1.7.2 Optimised Work Programme

After performing the budget optimisation, a revised work programme is produced. This has a total cost that is within the specified budget. The road sections are listed chronologically (by year) in accordance with the incremental benefit/cost ranking index described in Chapter B2.



1.7.3 Generate Reports

Several reports are included within the Programme Analysis application. For this case study, the **optimised work programme** report has been reproduced for this document (see Figure B1.2).

HDM-4 Optimised Work Programme

Study Name: **1. Western Province (Life Cycle Analysis)**

Run Date: **08-12-1999**

All costs are expressed in: US Dollar (millions)

Year	Section	Length (km)	AADT	Surface Class	Work Description	NPV/Cost	Financial Costs	Cumulative Costs
2000	MTS 549 km 0 - 1.0	1.0	2521	Bituminous	Overlay 50mm	3.38	0.13	0.13
2000	MSC 112 km 35 - 80	21.0	1770	Bituminous	Overlay 50mm	1.85	2.65	2.77
2000	MAN 203 km 254 - 262.8	8.8	1781	Bituminous	Overlay 50mm	1.83	1.13	3.91
2000	MSC 141 km 28 - 49.5	21.5	2154	Bituminous	Resealing SBS	1.28	0.92	4.83
2000	MSC 131 km 16.5 - 26	9.5	2154	Bituminous	Resealing SBS	1.14	0.42	5.25
2000	MTS 754 km 0 - 1.8	1.8	1074	Bituminous	Overlay 50mm	1.11	0.25	5.50
2000	MSW 959 km 0.1- 5, 10- 15.4	10.3	1265	Bituminous	Overlay 50mm	1.10	1.26	6.76
2000	MSE 203 km 80 - 90	10.0	1191	Bituminous	Overlay 50mm	0.76	1.26	8.02
2000	MAN 409 km 105 - 110	5.0	1076	Bituminous	Overlay 50mm	0.60	0.63	8.65
2000	MTN 836 km 0 - 13	13.0	926	Bituminous	Overlay 50mm	0.60	1.64	10.29
2000	MAN 243 km 0 - 21	21.0	1044	Bituminous	Overlay 50mm	0.53	2.65	12.94
2000	MTN 805 km 145 - 152.6	7.6	873	Bituminous	Overlay 50mm	0.41	1.01	13.94
2000	MSC 138 km 5 - 10 & 65 - 70	10.0	1315	Bituminous	Resealing SBS	0.21	0.42	14.36

This report was generated by version 1.0 of HDM-4.

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Figure B1.2 HDM-4 Optimised Work Programme

Appendix B2 - Programme Analysis Case Study 2

1 Eastern Province rehabilitation programme – Multi-Year Forward Programme

1.1 Introduction

This Case Study demonstrates the application of Programme Analysis to the production of a rehabilitation programme for a road network in the Eastern Province of a country. The study demonstrates the Multi-Year Forward Programme method described in Chapter B2.

The objective is again to prepare a prioritised list road projects from a selected group of road sections that have been identified to be in poor condition. In practice, this would follow some condition survey of the road network and the application of pavement condition trigger levels or thresholds to identify those sections in need of some form of rehabilitation.

This Case Study demonstrates the application of the Multi-year Forward Programme method for the production of a rehabilitation programme for a road network. The only difference with the life cycle analysis data set is the way that the maintenance standards are assigned to road sections. The multi-year forward programme method requires one maintenance standard to be assigned to each road section, while life-cycle analysis requires at least two standards per section to permit economic analysis. The multi-year forward programme method assigns a default do minimum alternative which corresponds to deferring capital road works (periodic maintenance and improvements) to the first year after the budget period.

To demonstrate the multi-year forward programme method, the Case Study is presented in the following steps:

- Step 1 : Overview of the case study data.
- Step 2 : Review of the case study data.
- Step 3 : Run multi-year forward programme analysis
- Step 4: Examine the analysis results

1.2 Overview of the case study data

The input data for this Case Study is labelled as:

Eastern Province Rehabilitation (Forward programme)

When a Programme Analysis study is created for the Multi-year forward programme method, the following data sets must be defined:

- Road network containing the candidate road sections for periodic maintenance and/or improvement.
- Vehicle fleet defining the characteristics of typical vehicles commonly found on the road network.
- The average annual daily traffic (AADT) on each road section.
- Maintenance and/or improvement standards to be assigned to each candidate road section.

The above data will usually be pre-defined in HDM-4 before a Programme Analysis study is created. The following data managers are used for this:

- **Road Network manager** - to create and edit the road section data
- **Vehicle Fleet manager** - to create and edit vehicle characteristics
- **Maintenance and Improvement standards manager**

To review the case study data, open the Programmes folder and select the **Eastern Province Rehabilitation (Forward Programme)** study as illustrated in Figure B2.1.

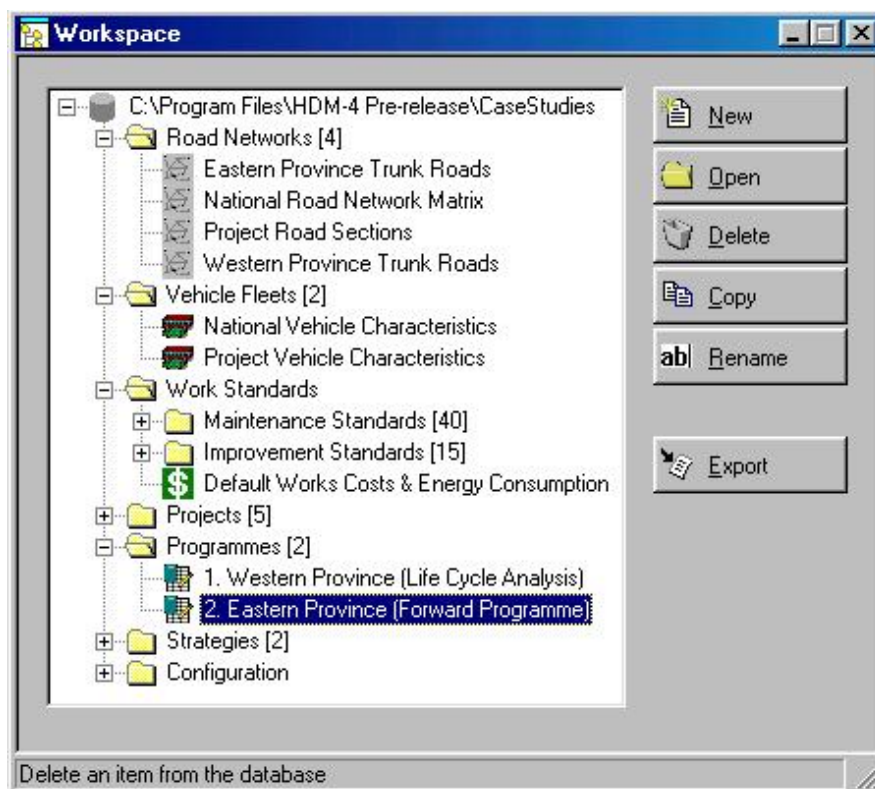


Figure B2.1 Workspace screen

The data used for this Case Study has already been created and therefore the procedure described below does not include the creation of a new study.

The procedure is similar to that of the Life Cycle method and involves defining details of the programme analysis to be carried out, assignment of work standards, generation of the work programme, optimisation under budget constraints, and finally generation of reports. This procedure should normally be followed sequentially, although iteration between the above tasks can be done in order to review previously specified data.

The data is arranged in groups (corresponding to the tasks) and, within these, the data are presented in different Tabs as summarised below:

1.2.1 Define Programme Details

- General
- Select Sections
- Select Vehicles
- Define Traffic

1.2.2 Assign Standards

- Work Programming

1.2.3 Generate Programme

- Perform Run
- Unconstrained Programme

1.2.4 Perform Budget Optimisation

- Define Budget
- Optimised Programme

1.2.5 Generate Reports

- Reports Tree

1.3 Multi-year forward programme data

The Case Study run data is held in the **Programmes** folder under:

Eastern Province Rehabilitation (Forward programme)

The data can be reviewed under the HDM-4 Programme Analysis Tabs listed below.

1.3.1 Define Programme Details

General

The **General** Tab can be used to define the study description, analysis type, analysis duration, the currency conversion rates and the discount rate to be used in the economic analysis. This Tab also shows the selected **Road Network** and **Vehicle Fleet**.

The analysis will be performed using the Multi-Year Forward Planning option based on a three-year budget period from 2000-2002.

Select Sections

The road network under study is represented by 21 sections (one section with unsealed surface class and all others with bituminous surfacing).

It is assumed that the long list of candidate road sections from Eastern Province was drawn up following some pavement condition survey. The list would normally contain only those road sections deemed to require some form of periodic maintenance or rehabilitation during the next budget period. Consequently, all road sections have been selected for the analysis.

However, if the road network had contained details of all road sections in Eastern Province, it is possible to select only those that should be included in the work programme for the next budget period. Alternatively, all road sections in the database could be analysed, particularly if this involves a long budget period. The selected sections define a sub-network for which a prioritised programme is required.

It is recommended that work programmes should be prepared for short-term budget periods, for example less than five years.

Details of each road section can be edited from within this Tab (press the **View/Edit Network** button).

Select Vehicles

The vehicles selected from the pre-defined Provincial Vehicle Characteristics fleet are indicated below. Individual vehicle attributes may be reviewed by double-clicking on the appropriate vehicle type description (or by selecting the section and clicking the **Edit Vehicle** button).

Define Traffic

This Tab defines the traffic composition and growth rates on the selected road sections. The initial traffic composition and growth rates (by vehicle type) can be reviewed by double-clicking the section description (or by selecting the section and clicking the **Section Traffic Details** button).

This Tab specified the traffic composition and annual growth rates that are applicable on each road section for each vehicle type in this analysis. Note that the traffic details can be copied from one road section to others, thereby permitting each road section to have a different road set of traffic growth characteristics, if required.

1.3.2 Assign Standards

Work programming

For Multi-Year Forward Programme, one maintenance standard (representing the rehabilitation alternative) is assigned to each road section, or, if an improvement is proposed, this is assigned together with associated maintenance standards before and after the improvement is applied. For Multi-Year Forward Programme, the base alternative is automatically assigned as the works needed in the first year following the defined budget period (with no works during the budget period).

In this Case Study, one maintenance policy has been assigned to all paved road sections and one to unsealed sections based on the Surface Class (that is, bituminous or unsealed) as follows:

Bituminous sections:  P&CS, Reseal, Overlay & Reconstruct (short code Brehab)
where P&CS = Patching and Crack Sealing

Unsealed section:  Grade, Spot regravels & Resurface (short code Grehab)

Each maintenance standard comprises a set of condition responsive works activities and establishes a maintenance policy for the road sections to which the standard is assigned. For this Case Study, one standard has been set up for each surface class. In practice this may be further refined by assigning different maintenance standards based on, for example, road class or traffic level, both of which indicate the relative importance of the section..

Details of the works activities and condition responsive intervention criteria associated with each maintenance standard are summarised in Table B2.1 and Table B2.2. The details may be reviewed under **Maintenance Standards** in **HDM-4 Workspace** shown in Figure B2.1.

Programme: 2. Eastern Province (Forward Programme)

Work Programming

Section	Surface class	Road class	Maintenance	Improvement	Maintenance after Improvement
Francis Town - Selebi P	Bituminous	Primary or Trunk	Brehab		
Gaborone-Lobatse Sect	Bituminous	Primary or Trunk	Brehab		
Gaborone-Pilane	Bituminous	Primary or Trunk	Brehab		
Gaborone-Lobatse Sect	Bituminous	Primary or Trunk	Brehab		
Gaborone-Tlokweng	Bituminous	Primary or Trunk	Brehab		
Jwaneng-Sekoma Secti	Bituminous	Secondary or Mair	Brehab		
Gravel Road	Unsealed	Tertiary or Local	Grehab		
Jwaneng-Sekoma Secti	Bituminous	Secondary or Mair	Brehab		

Assign standard...

Save

Close

Assignment of works to selected sections

Paved Road Maintenance Standard

This standard (P&CS, Reseal, Overlay & Reconstruct - Brehab), adopted for bituminous sections, includes five maintenance works items defined in Table B2.1 below:

Table B2.1
Details of Maintenance Standard assigned to bituminous road sections
short code: Brehab

Works Type	Works Activity	Intervention Criteria	Ranking *
Routine Pavement	Patching	Potholes ≥ 10 per km	22
	Crack sealing	Transverse cracks ≥ 15 per km	22
Resurfacing	Surface Dressing Single	Total damaged area $\geq 30\%$	16
Rehabilitation	Overlay dense graded asphalt	Roughness ≥ 5 IRI AND Cracking $\geq 5\%$	9
Reconstruction	Pavement Reconstruction	Roughness ≥ 12 IRI	6

Note: * Ranking of road works from Chapter D2 [Analytical Framework and Model Descriptions](#).

If more than one of these activities are triggered in a given year, the works item that is highest in the hierarchy of road works (that is, with lowest ranking value) will be applied. Note that the routine pavement works (patching and crack sealing) have the same ranking, and both can be applied in the same maintenance year. (The hierarchy is defined by the ranking of road works in Chapter D2 of [Analytical Framework and Model Descriptions](#) - Modelling Road Works Effects).

Unsealed Maintenance Standard

The Works Activities that make up this standard for unsealed road sections (Grade, Spot Regravell & Resurface - Grehab) are as follows.

Table B2.2
Details of Maintenance Standard assigned to unsealed road sections
short code: Grehab

Works Type	Works Activity	Intervention Criteria	Ranking *
Routine Pavement	Grading	Roughness ≥ 8 IRI	7
	Spot regravelling	Gravel thickness ≤ 100 mm	7
Resurfacing	Regravelling	Gravel thickness ≤ 50 mm	6

Note: * Ranking of road works from Chapter D2 of [Analytical Framework and Model Descriptions](#)

The ranking order assigned to these road work items implies that Resurfacing has the highest priority. Spot regravelling and Grading have equal ranking, and therefore both of these can be performed in a given analysis year.

Details of the maintenance works items can be reviewed under **Maintenance Standards** in **HDM-4 Workspace** shown in Figure B2.1.

1.4 Generate Programme

Perform Run

For Multi-Year Forward Programme Analysis, the user can choose to perform the run either with or without economic analysis. Economic analysis is required before budget optimisation. This can be selected using the **Run Setup** button.

The **Run Setup** button also allows the user to select Energy Balance and Emissions models, to model acceleration effects, and to include accident costs in the economic analysis.

When **Run Setup** is completed, the Work Programme (prior to budget optimisation) is generated by pressing the **Start** button. Once completed, the user is strongly advised to **Save** the generated programme, especially before viewing any reports.

Unconstrained Programme

The generated Work Programme can be viewed on screen prior to performing a budget optimisation.

Programme: 2. Eastern Province (Forward Programme)

Perform Run | Unconstrained Programme

Budget Period Analysis - performed at 08-12-99

Road Section	Road class	Length	MT AADT	Road Works	Year	Cost (m#)	Capital Cum. Cost	NPV/C
Mosetse-Nata	Secondary o	68.0	177	Prep. Edge Repair	2000	0.0000	-	3.00
Palapye-Serule	Primary or Tr	70.0	963	Overlay 50mm	2000	8.4420	63.4	0.77
Palapye-Serule	Primary or Tr	70.0	963	Prep. Patching	2000	0.0000	-	0.77
Palapye-Serule	Primary or Tr	70.0	963	Prep. Edge Repair	2000	0.0001	-	0.77
Pilene-Ratesa	Secondary o	7.0	459	Resealing SBSD	2000	0.2814	63.7	3.38
Pilene-Ratesa	Secondary o	7.0	459	Prep. Edge Repair	2000	0.0000	-	3.38
Ratesa-Monametsana	Secondary o	15.0	249	Patching	2000	0.1266	-	0.22
Sebina - Mosetse	Secondary o	79.0	202	Resealing SBSD	2000	3.1758	66.9	1.53
Sebina - Mosetse	Secondary o	79.0	202	Prep. Patching	2000	0.0000	-	1.53
Sebina - Mosetse	Secondary o	79.0	202	Prep. Edge Repair	2000	0.0001	-	1.53
Sekoma-Monwamosu	Secondary o	79.0	313	Overlay 50mm	2000	8.5320	75.4	0.17
Sekoma-Monwamosu	Secondary o	79.0	313	Prep. Edge Repair	2000	0.0000	-	0.17
Serule-Francis Town	Primary or Tr	98.0	1119	Resealing SBSD	2000	3.9396	79.3	3.08
Serule-Francis Town	Primary or Tr	98.0	1119	Prep. Patching	2000	0.0000	-	3.08

Manually select work items prior to optimisation

1.5 Perform Budget Optimisation

Define Budget

The budget optimisation provides a facility to select road sections that can be included within a specified budget in order to maximise the economic benefits. Given that the selected road sections from Eastern Province would require US\$ 80 million in over the three-year period from 2000 to 2002 for periodic maintenance, it is now possible to select the optimal set of road sections if less than 100% of the required budget is available. Assume that the amount of money available for the three-year period has been set at US\$ 60 million.

Optimised Programme

After performing the budget optimisation, a revised work programme is produced. This has a total cost that is within the specified budget. The road sections are listed chronologically (by year) in accordance with the incremental benefit/cost ranking index described in Chapter B2.

Table B2.3 Details of alternatives for life-cycle analysis

Alternative	Surface Class	Maintenance Standard	Effective from year	Maintenance Works
Base	Bituminous	Patching and Crack Sealing (B/rout)	1999	Pothole Patching (PATPOT)
				Crack Sealing (CRKSL)
	Unsealed	Grade & Spot Regravel (G/rout)	1999	Grading (GRADE)
				Spot Regravel (SPRG)
Rehabilitation	Bituminous	P&CS, Reseal, Overlay & Reconstruct (Brehab)	1999	Pothole Patching (PATPOT) *
				Crack Sealing (CRKSL) *
				Reseal (RSL30)
				Overlay (OV5IRI)
				Reconstruct (REC12)
	Unsealed	Grade, Spot Regravel & Resurface (Grehab)	1999	Grading (GRADE) *
				Spot Regravel (SPRG) *
				Resurface (RESURF)

Notes:

- 1 Details of intervention criteria for works activities are given in Table B2.1 and Table B2.2.
- 2 Routine pavement works in rehabilitation alternative (identified by *) are identical to those in base alternative.

HDM-4 Unconstrained Work Programme

Study Name: **2. Eastern Province (Forward Programme)**

Run Date: **08-12-1999**

All costs are expressed in: US Dollar (millions)

Year	Section	Road Class	Length (km)	AADT	Surface Class	Work Description	NPV/C	Financial Costs	Cumulative Costs
2000	Gaborone-Lobatse Section 2	Primary or Trunk	34.0	1334	Bituminous	Resealing SBS	128.46	1.37	1.37
2000	Gaborone-Pilane	Primary or Trunk	30.0	1516	Bituminous	Resealing SBS	4.06	1.21	2.57
2000	Pilane-Rasesa	Secondary or Main	7.0	459	Bituminous	Resealing SBS	3.38	0.28	2.85
2000	Gaborone-Lobatse Section 1	Primary or Trunk	60.0	1814	Bituminous	Resealing SBS	3.34	2.41	5.27
2000	Serule-Francis Town	Primary or Trunk	98.0	1119	Bituminous	Resealing SBS	3.08	3.94	9.21
2000	Mosetse-Nata	Secondary or Main	68.0	177	Bituminous	Resealing SBS	3.00	2.73	11.94
2000	Gaborone-Tlokweng	Primary or Trunk	10.0	2586	Bituminous	Overlay 50mm	1.63	1.21	13.15
2000	Sebina - Mosetse	Secondary or Main	79.0	202	Bituminous	Resealing SBS	1.53	3.18	16.32
2000	Francis Town - Selebi Phikwe	Primary or Trunk	85.0	1226	Bituminous	Resealing SBS	1.33	3.42	19.74
2000	Palapye-Serule	Primary or Trunk	70.0	963	Bituminous	Overlay 50mm	0.77	8.44	28.18
2000	Lobatse-Jwaneng Section 1	Primary or Trunk	120.0	812	Bituminous	Overlay 50mm	0.38	15.55	43.73
2000	Jwaneng-Sekoma Section 1	Secondary or Main	80.0	607	Bituminous	Overlay 50mm	0.34	9.65	53.38
2000	Monametsana-Mahalapye	Secondary or Main	142.0	446	Bituminous	Overlay 50mm	0.33	17.13	70.51
2000	Gravel Road	Tertiary or Local	10.0	114	Unsealed	Resurface	0.18	0.29	70.80
2000	Sekoma-Morwamosu	Secondary or Main	79.0	313	Bituminous	Overlay 50mm	0.17	8.53	79.33
2002	Rasesa-Monametsana	Secondary or Main	15.0	270	Bituminous	Resealing SBS	0.22	0.60	79.93

This report was generated by version 1.0 of HDM-4.

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Figure B2.2 HDM-4 Unconstrained Work Programme